

Software Requirement Specification (SRS) For Security Incident Management (SIM) System

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the localized Security Incident Management (SIM) platform. It provides a clear, structural breakdown of the features and performance metrics required for the application's implementation and evaluation.

1.2 Scope

High rates of campus and community crime, driven by factors such as cultism, drug abuse, poor environmental lighting, and slow manual reporting processes, present a significant safety challenge. The SIM platform addresses these challenges by introducing a centralized digital reporting channel. Please note that this project is developed as a mini-prototype system and is not a full-scale, production-ready deployment. It serves to demonstrate core concepts of crowd-sourced security logging and localized incident mapping.

1.3 Intended Audience

This document is prepared for students, staff, community landlords, and the course instructor to evaluate the system design and technical feasibility.

2. Overall Description

2.1 Product Perspective

"SIM is a standalone web-based application accessible via any browser." It interfaces directly with a database to store submitted reports and update a public incident visualization interface without requiring native device installations.

2.2 User Classes

- Student: Campus-based individuals who need a quick, friction-free interface to report incidents anonymously and check safe routes.
- Staff: University personnel who utilize the system to report office or perimeter security breaches and monitor community safety alerts.
- Landlord: Off-campus residential owners who report neighborhood crimes and monitor threat levels around student housing zones.
- Admin (Campus Security Personnel): Institutional security officers who log in to verify incoming reports, update incident statuses, and manage the live threat map.

3. Requirement Gathering Summary



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Untitled form

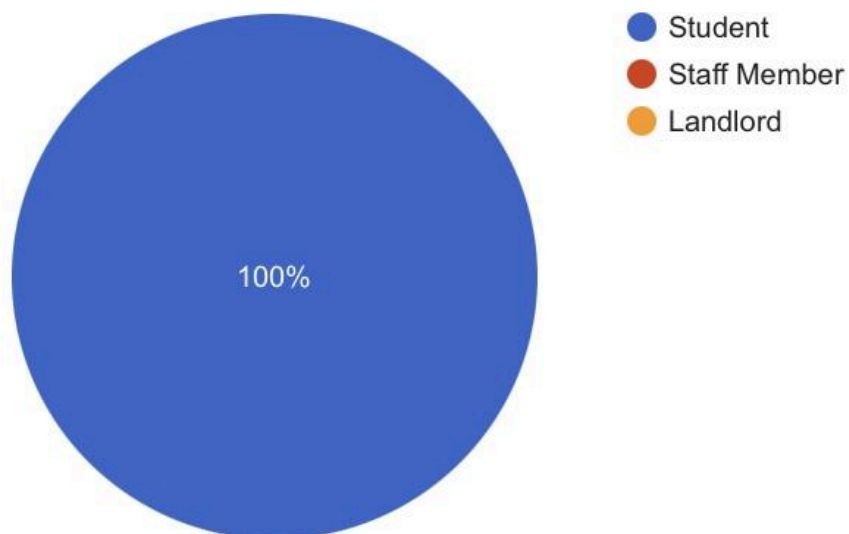
Questions

Responses

4

Settings

4 responses



Which of these security incidents have you witnessed or experienced on or around campus?(Select all that apply)

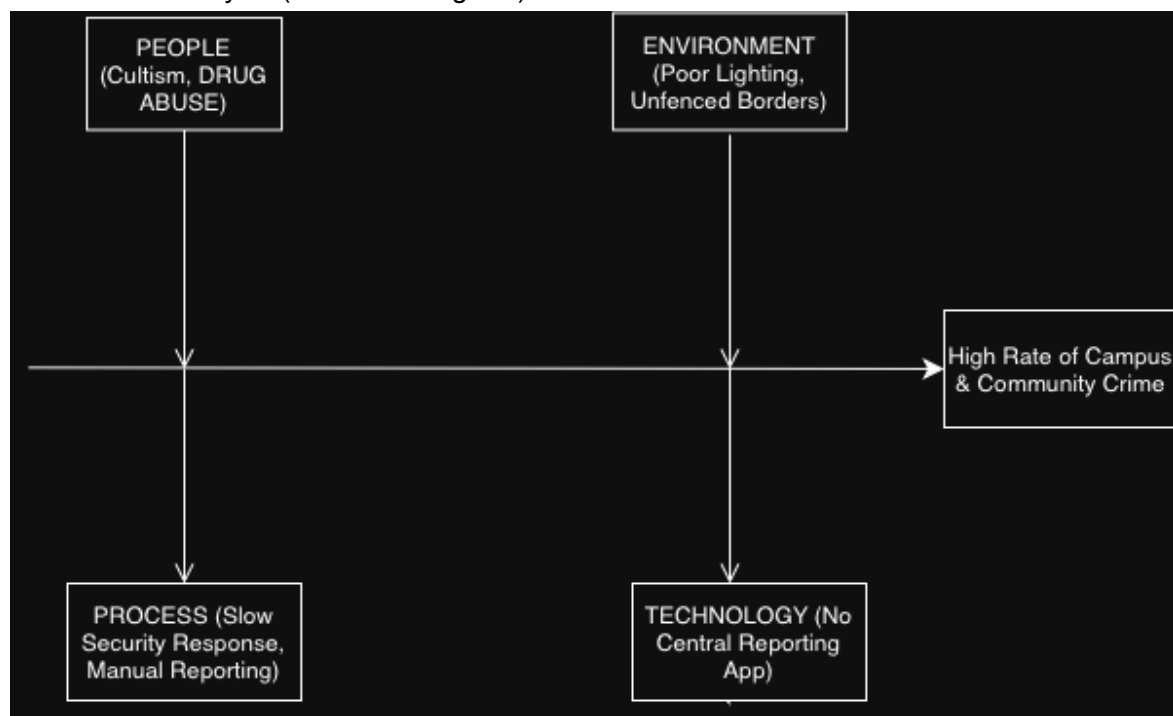


Copy chart

4 responses

Data Summary: Primary data was gathered using a targeted digital questionnaire distributed within the community. Initial metrics show that 100% of the respondents identifying as students face localized security threats. Among these, 75% have directly witnessed or experienced incidents of Theft/Stealing, 50% have encountered Physical Assault/Stabbing, and 25% reported instances of Sexual Assault. This localized sample clearly indicates that property theft and physical violence are the prevailing threats, validating the critical need for an immediate, responsive digital tracking platform to map occurrences and improve situational awareness.

Root-Cause Analysis (Ishikawa Diagram)



Our root-cause analysis revealed that the absence of a centralized digital reporting platform, combined with an unclear or slow manual reporting policy, serves as a major contributor to crime underreporting. Furthermore, environmental vulnerabilities like poor lighting and unfenced borders exacerbate these systemic issues, highlighting the critical need for an immediate technological solution to bridge the communication gap.

4. Functional Requirements

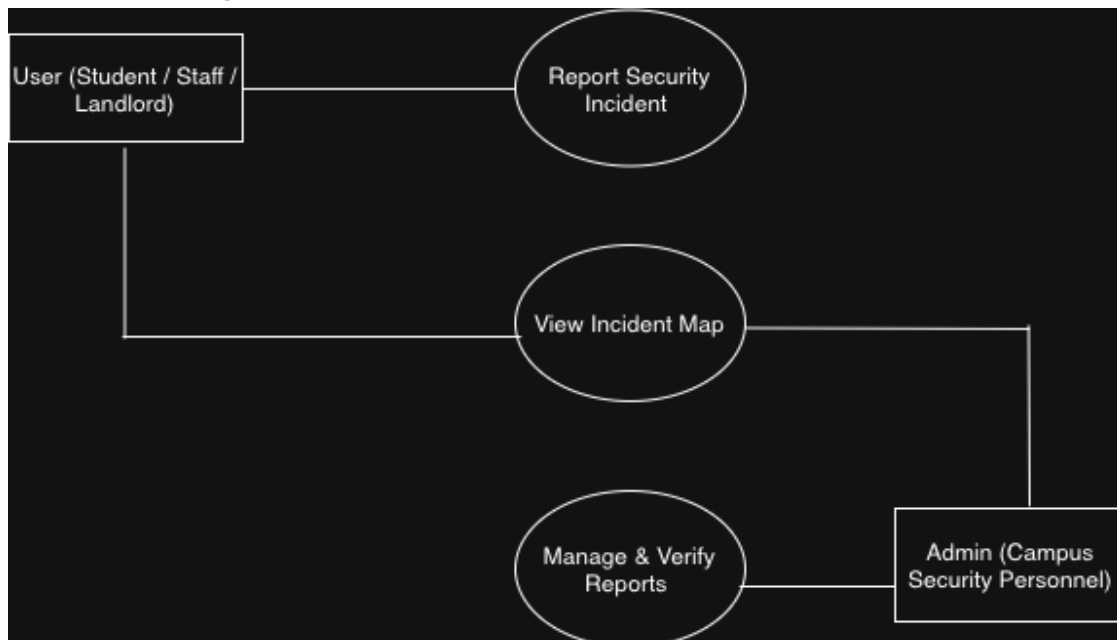
- FR1: The system shall allow users to submit an incident report including the specific crime type, location details, and a clear text description.
- FR2: The system shall allow users to submit safety reports completely anonymously to protect user privacy.

- FR3: The system shall generate a unique reference ID automatically upon every successful report submission.
- FR4: The system shall display an immediate confirmation message to the user on the screen after a report is sent.
- FR5: The system shall provide a secure login interface restricted to authorized Admin users (Campus Security Personnel).
- FR6: The system shall allow Admin users to view a dashboard of pending reports and switch their status to verified or archived.

5. Non-Functional Requirements

- NFR1 (Usability): The interface shall be simple enough for non-technical users to operate efficiently without any prior technical training. To ensure accessibility for all individuals, the styling must utilize a functional, high-contrast theme focused on neutral elements rather than gendered design choices.
- NFR2 (Availability): The system shall be accessible via any standard mobile or desktop web browser at any time without requiring external plugins or app store downloads.
- NFR3 (Security/Privacy): Anonymous incident reports shall not collect, log, or store any identifying user information, IP addresses, or device data in the database.
- NFR4 (Performance): Form submission actions shall complete processing and successfully display the on-screen confirmation message within 2 seconds of clicking submit.

6. Use-Case Diagram



Actor Descriptions

- User (Student / Staff / Landlord): This actor can submit localized safety incident details completely anonymously and view the collective interactive incident map to stay informed on high-risk zones.
- Admin (Campus Security Personnel): This actor can securely log into the system management panel to review pending alerts, verify incoming citizen reports, and manage the live threat map.

7. Object Listing

This section outlines the primary data objects tracked within the prototype system to handle incident records and user sessions:

Object Name: IncidentReport

Attributes / Fields: reportId, crimeType, locationDetails, descriptionText, timestamp, status

Description: Stores the crowd-sourced data payload for every submitted crime report.

Object Name: AdminUser

Attributes / Fields: adminId, username, passwordHash, role

Description: Holds credentials and authorization levels for campus security dispatchers.

Object Name: MapMarker

Attributes / Fields: markerId, associatedReportId, coordinates, riskLevel

Description: Represents the spatial visual coordinate data rendered on the public view interface.

8. System Architecture

SIM is built using HTML, CSS, and JavaScript for the frontend layout, and is hosted directly using GitHub Pages. To keep the setup efficient and accessible during this initial development phase, the current prototype utilizes browser-based local storage (localStorage) to handle form data processing. A full future production version of the platform would scale up to integrate a relational database management system, such as MySQL or a cloud solution like Firebase, to safely persist and sync reports across all network clients.

9. Conclusion

The Security Incident Management (SIM) system successfully provides a localized, centralized digital workspace to combat campus and community crime underreporting. By offering an alternative to slow, manual tracking loops through simple web-based reporting and a public visual map, the application vastly improves response visibility and awareness. Ultimately, this mini-prototype demonstrates how a targeted web framework can bridge the communication gap between citizens and security operatives to create a safer environment.